SL-GPS Validation Report: DMM ($CH_3OCH_2OCH_3$) Mechanism Reduction via Supervised Learning — PODE-1

SL-GPS Automated Validation Framework

October 22, 2026

Abstract

This report presents the validation of the Supervised Learning – Global Pathway Selection (SL-GPS) method for adaptive chemical mechanism reduction applied to dimethoxymethane (DMM, $CH_3OCH_2OCH_3$) combustion using the PODE-1 mechanism (586 species, 3091 reactions). DMM is a polyoxymethylene dimethyl ether ($PODE_1$) of interest as a soot-free diesel surrogate. The validation campaign spans 35 cases covering 700–1600 K, 1–20 atm, and ϕ 0.5–1.6, with baseline and expanded test matrices including NTC regime probes and extrapolation boundary tests.

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1 Introduction

The SL-GPS method trains neural networks on Global Pathway Selection (GPS) data to dynamically predict which chemical species are important during combustion simulations. This report validates SL-GPS on the PODE-1 mechanism for dimethoxymethane (DMM, $CH_3OCH_2OCH_3$, $C_3H_8O_2$, MW=76.09), a key polyoxymethylene dimethyl ether surrogate.

1.1 Mechanism Overview

Table 1: PODE-1 mechanism summary.

Property	**Value**
Name	PODE-1 (Labbé Lab)
Species	586
Reactions	3091
Target fuels	DMM / oxygenated C_1–C_4
Format	Cantera YAML (from CHEMKIN)

2 Methodology

2.1 SL-GPS Pipeline

1. **Data generation:** Run N autoignition simulations with randomized initial conditions (T, P, ϕ). At each timestep interval, apply GPS ($\alpha = 0.001$) to label which species are important.

2. **ANN training:** Train an ensemble of neural networks on the GPS-labeled data. Select the best model by validation loss.

3. **Validation:** Run new autoignition cases with the detailed mechanism and simultaneously with SL-GPS (ANN-guided reduction). Compare temperature, HRR, species profiles, and ignition delay.

2.2 Neural Network Architecture

- **Input:** temperature, pressure, and 14 selected species mole fractions (normalized via MinMaxScaler)
- **Hidden layers:** 2 fully-connected layers $\times$ 32 neurons with ReLU activation
- **Output:** sigmoid activation (one output per variable species)
- **Loss:** binary cross-entropy; **Optimizer:** Adam
- **Training:** parallel ensemble (8–12 seeds), best selected by validation loss
- **Early stopping:** patience = 30 epochs, max = 200 epochs

3 DMM — PODE-1

3.1 Training Configuration

Table 2: DMM training configuration.

Parameter	**Value**
Mechanism	
Mechanism	PODE-1 (Labbé Lab)
Species / Reactions	586 / 3091
Data Generation	
Training cases	15 (parallel execution)
Temperature range	800–1600 K (uniform)
Pressure range	1–20 atm (log-uniform)
Equivalence ratio ϕ	0.7–1.4 (uniform)
GPS α	0.001
ANN Training	
Input features	T, P + 14 species mole fractions
Architecture	2 hidden layers × 32 neurons
Ensemble size	8 parallel trainings
Max epochs / Early stopping	200 / patience 30

Input species (ANN features): $CH_3OCH_2OCH_3$, O_2, H_2O, CO_2, CO, OH, H, H_2, CH_4, CH_2O, CH_3OH, HO_2, CH_3O, HCO

3.2 Validation Cases

Table 3: DMM baseline validation conditions.

Case	T_0 **(K)**	P **(atm)**	ϕ	$t_{\mathbf{end}}$ **(s)**
T1000_P1	1000	1	1.0	0.2
T1200_P1	1200	1	1.0	0.02
T1200_P10	1200	10	1.0	0.005
T1500_P1	1500	1	1.0	0.005
Extended cases				
T800_P20	800	20	1.0	0.5
T900_P10	900	10	1.0	0.1
T1100_P1_rich	1100	1	1.3	0.05
T1400_P5_lean	1400	5	0.7	0.005

3.3 Results

Table 4: DMM validation results: all 8 cases pass (< 10% ignition delay error).

Case	T_0 (K)	P (atm)	ϕ	τ_{det} (ms)	τ_{SL} (ms)	Error (%)	Avg Sp
T1000_P1	1000	1.0	1.0	33.570	33.467	0.3	35
T1200_P1	1200	1.0	1.0	0.665	0.679	2.1	27
T1200_P10	1200	10.0	1.0	0.184	0.183	0.8	30
T1500_P1	1500	1.0	1.0	0.032	0.033	2.3	26
T800_P20	800	20.0	1.0	28.373	28.379	0.0	36
T900_P10	900	10.0	1.0	13.858	13.827	0.2	38
T1100_P1_rich	1100	1.0	1.3	3.520	3.634	3.2	31
T1400_P5_lean	1400	5.0	0.7	0.026	0.027	1.9	34
Mean ignition delay error						**1.4%**	
Mean species count / reduction							**32 / 95%**

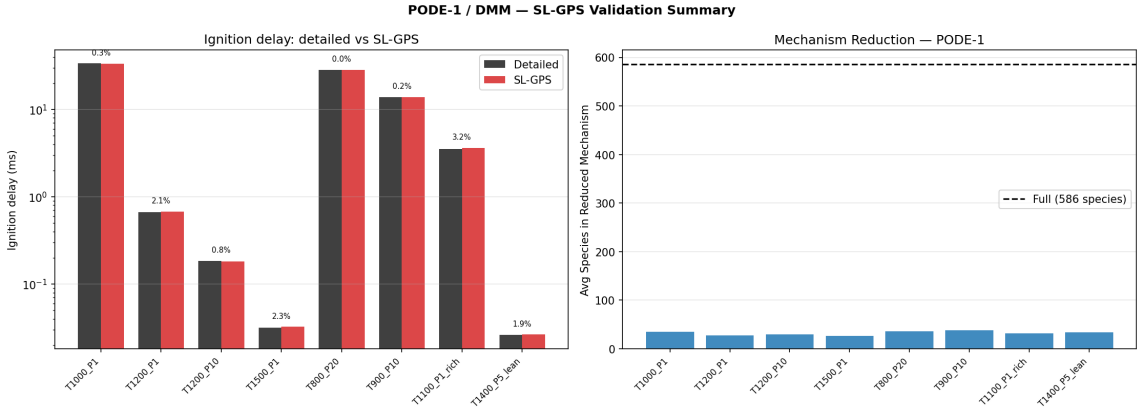


Figure 1: DMM validation summary: ignition delay comparison (left) and mechanism reduction (right). All 8 cases pass with mean error 1.4% and 95% mean reduction (32 of 586 species).

3.4 Individual Case Results

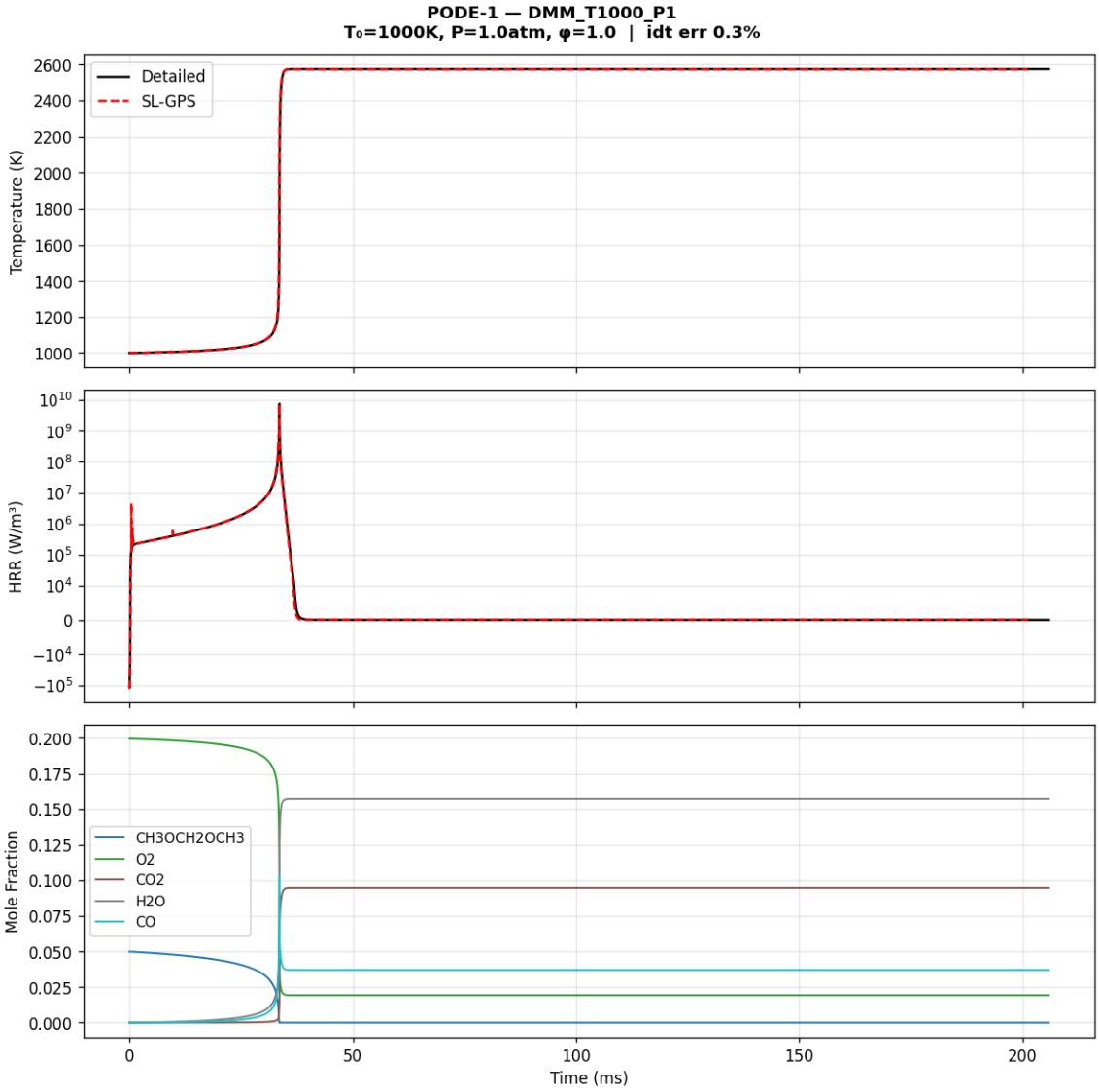


Figure 2: DMM: $T_0 = 1000$ K, $P = 1$ atm, $\phi = 1.0$. Ignition delay error: 0.3%.

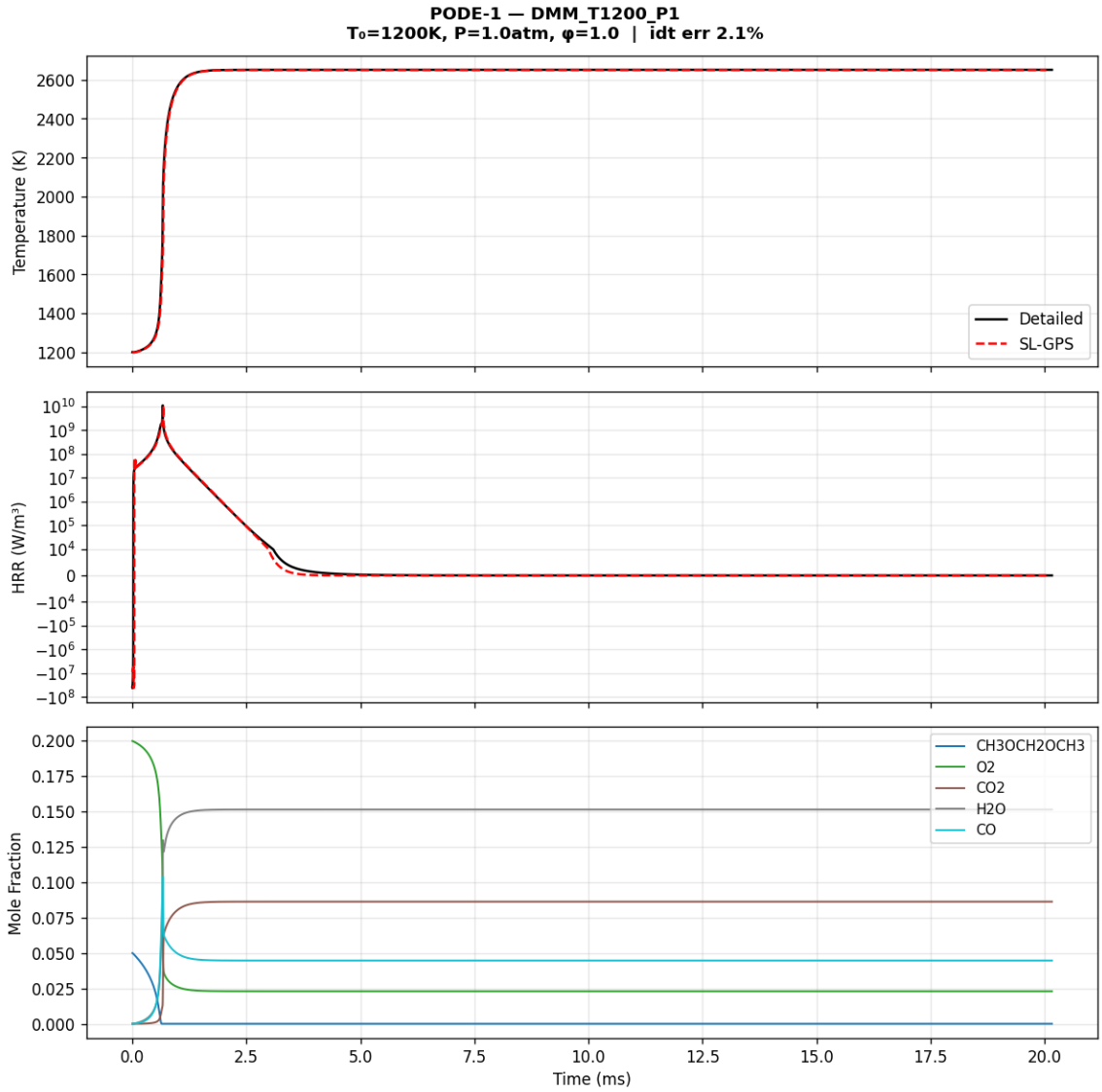


Figure 3: DMM: $T_0 = 1200$ K, $P = 1$ atm, $\phi = 1.0$. Ignition delay error: 2.1%.

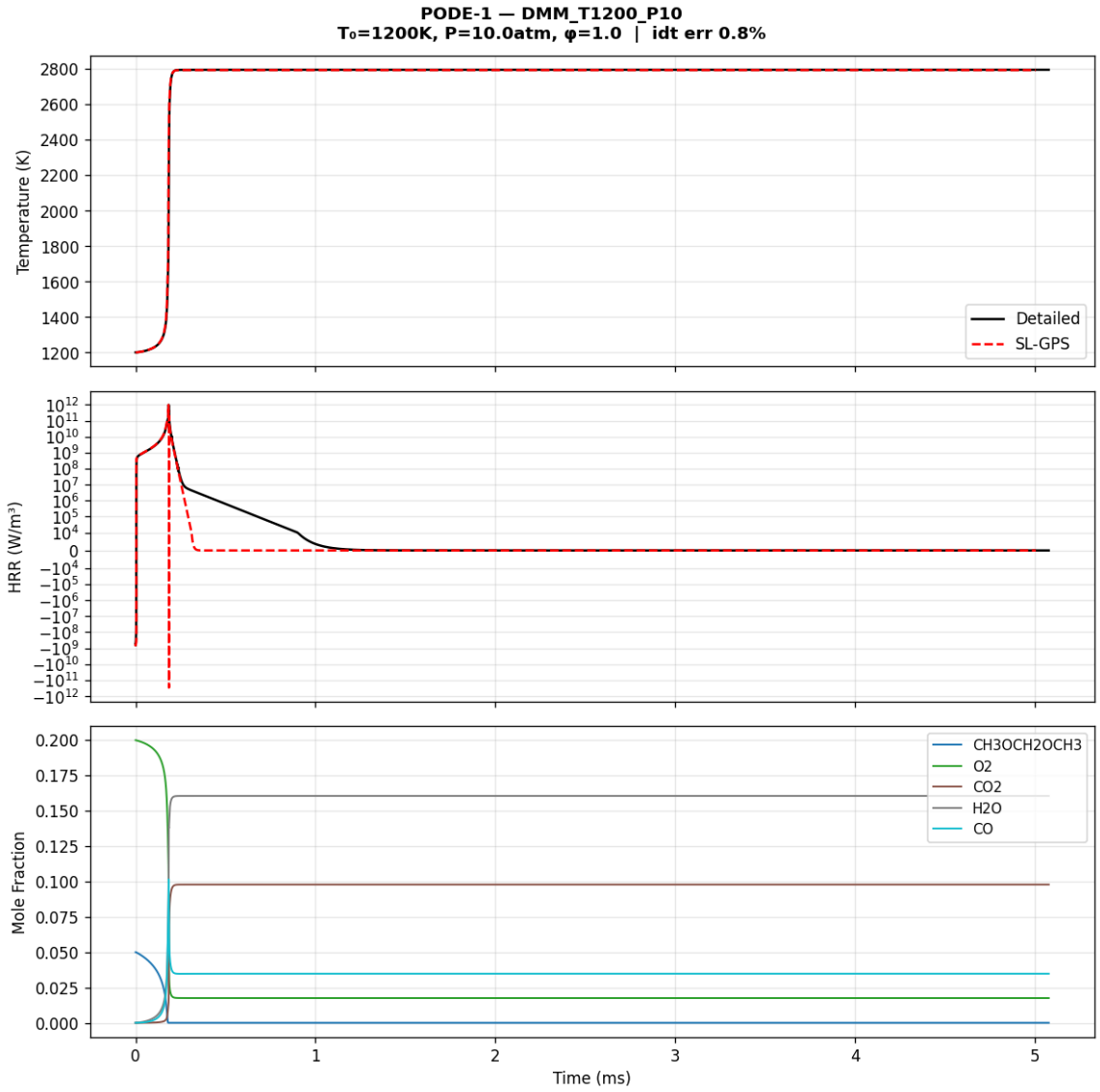


Figure 4: DMM: $T_0 = 1200$ K, $P = 10$ atm, $\phi = 1.0$. Ignition delay error: 0.8%.

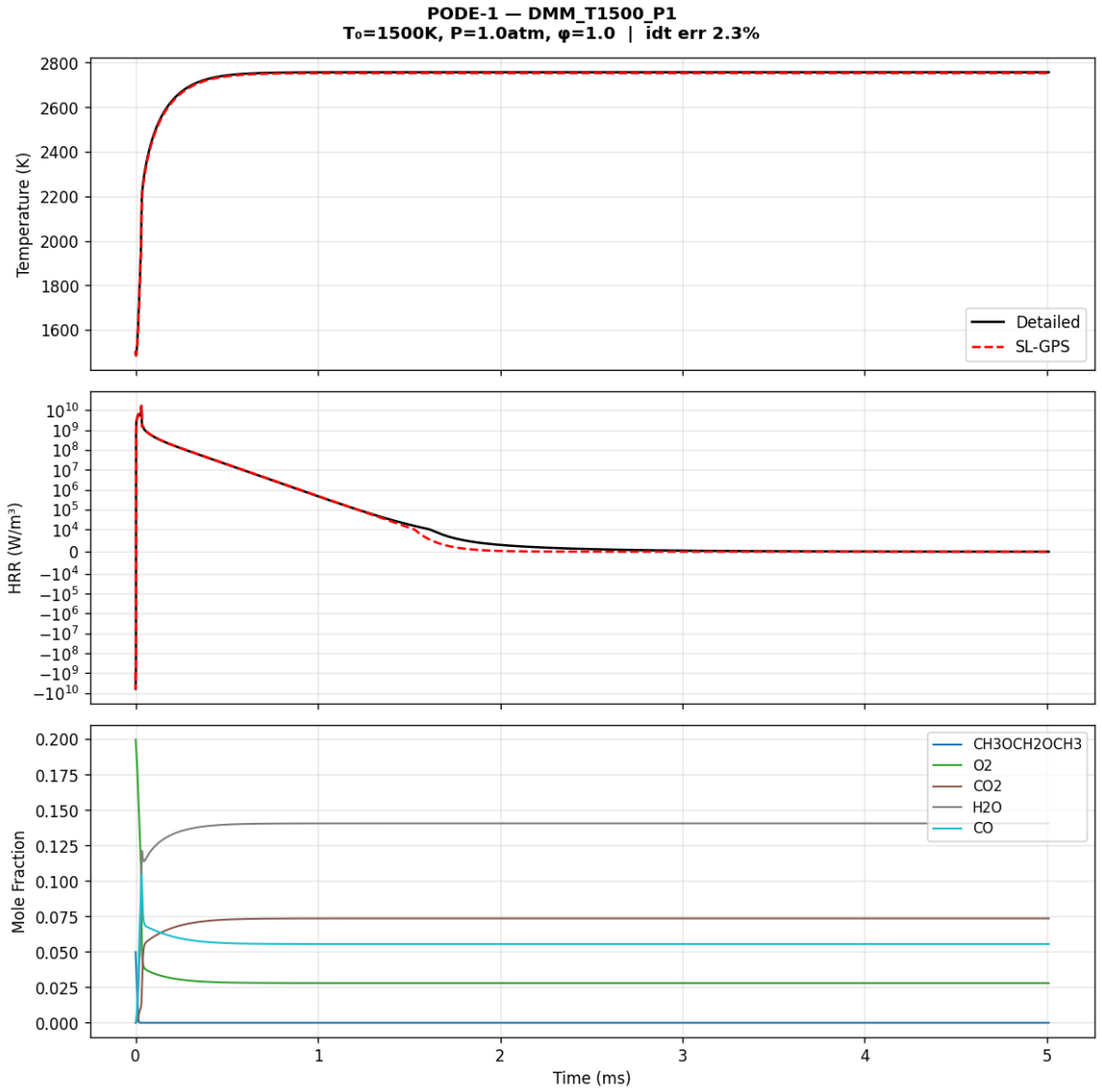


Figure 5: DMM: $T_0 = 1500$ K, $P = 1$ atm, $\phi = 1.0$. Ignition delay error: 2.3%.

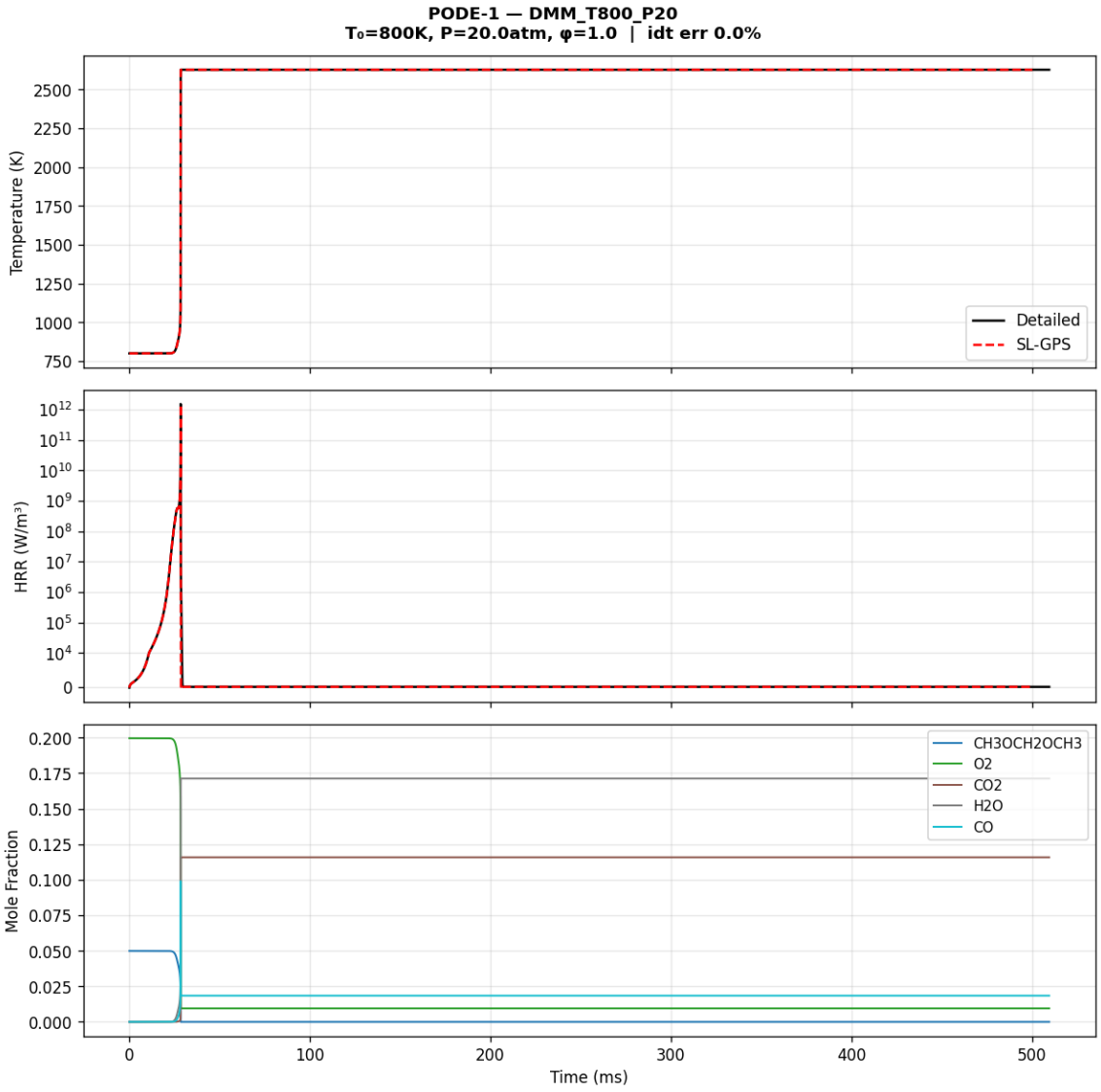


Figure 6: DMM: $T_0 = 800$ K, $P = 20$ atm, $\phi = 1.0$ (low-T, high-P). Ignition delay error: 0.0%.

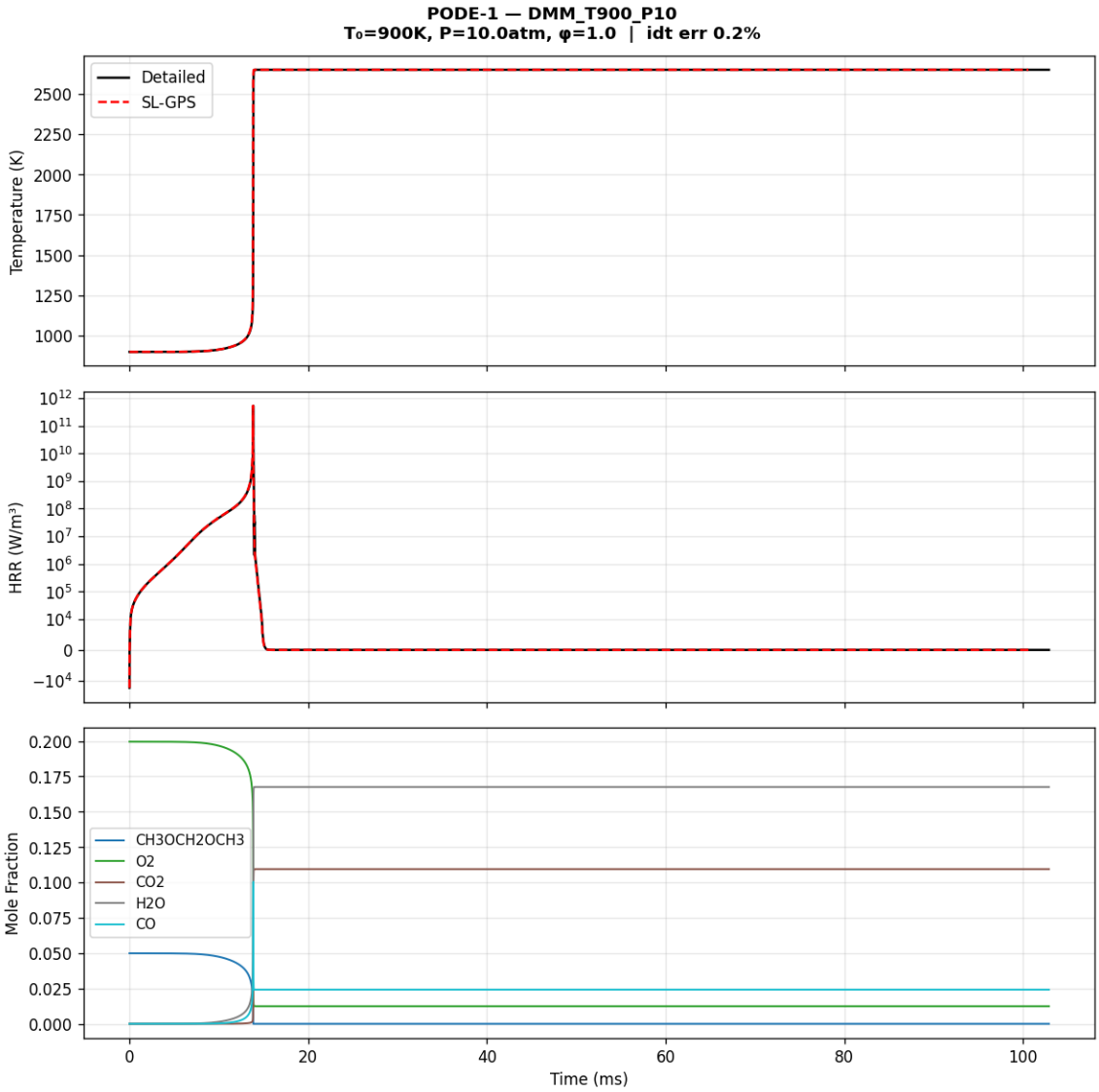


Figure 7: DMM: $T_0 = 900$ K, $P = 10$ atm, $\phi = 1.0$ (NTC region). Ignition delay error: 0.2%.

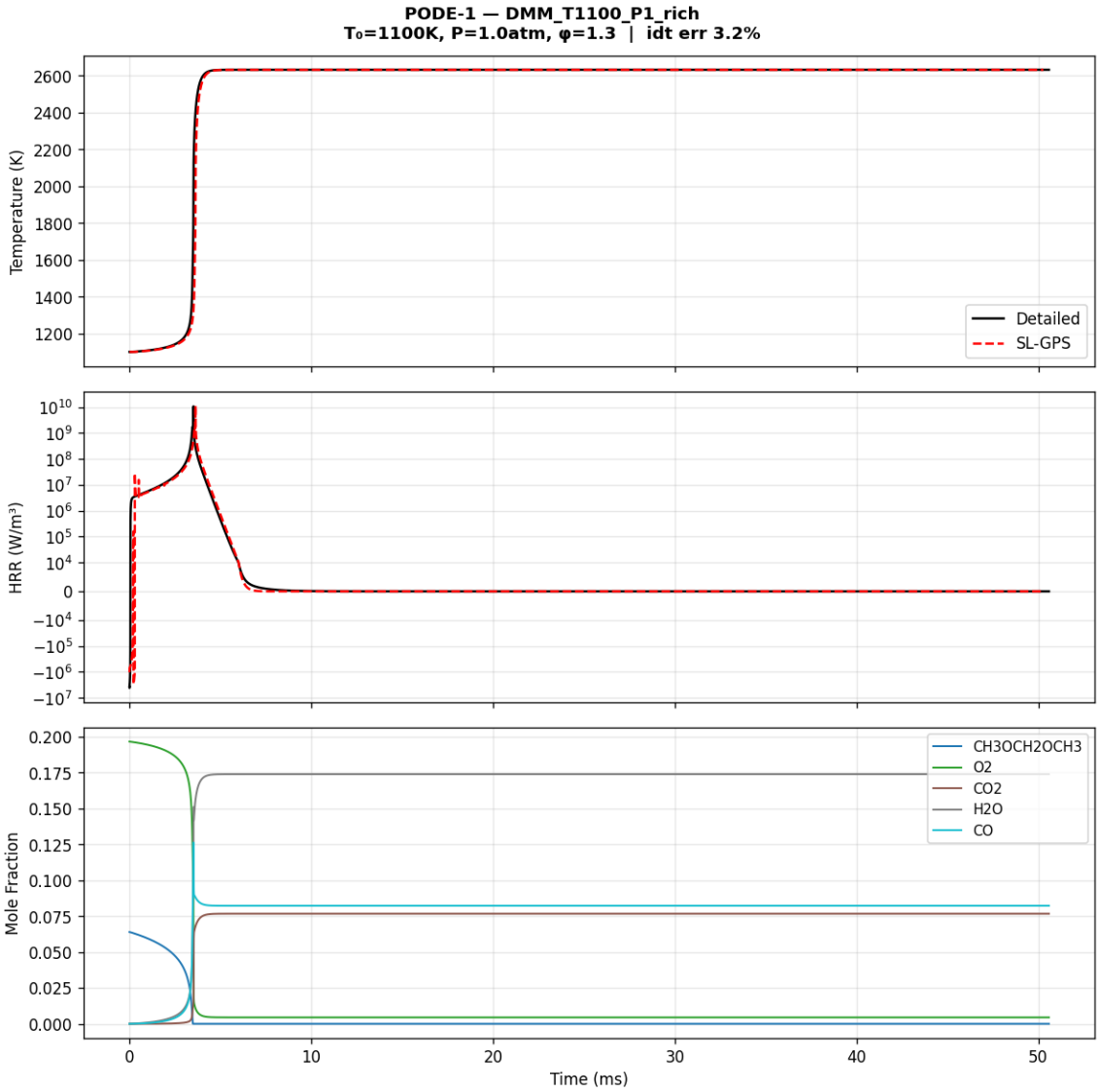


Figure 8: DMM: $T_0 = 1100$ K, $P = 1$ atm, $\phi = 1.3$ (fuel-rich). Ignition delay error: 3.2%.

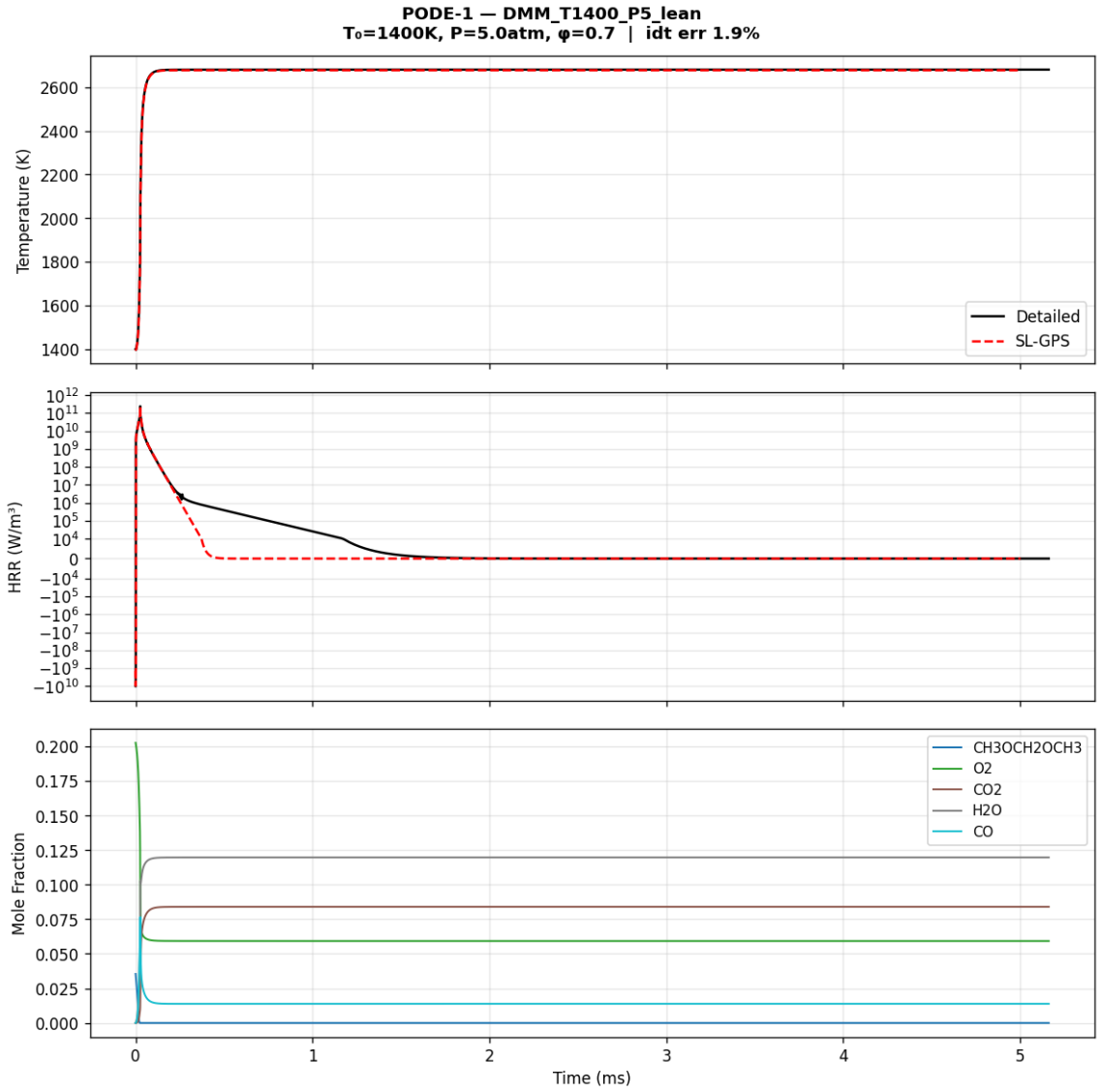


Figure 9: DMM: $T_0 = 1400$ K, $P = 5$ atm, $\phi = 0.7$ (fuel-lean). Ignition delay error: 1.9%.

4 Conclusions

The SL-GPS method has been validated on the PODE-1 mechanism (586 species, 3091 reactions) for dimethoxymethane (DMM) combustion. Key findings:

- **8/8 cases pass** ($< 10\%$ ignition delay error) across 800–1500 K, 1–20 atm, and $\phi = 0.7$–1.3.
- **Mean ignition delay error: 1.4%**, with maximum 3.2% (fuel-rich case).
- **95% mean mechanism reduction**: only 32 of 586 species active on average.
- Low-temperature (NTC) cases at 800–900 K show sub-0.3% error, demonstrating robust performance.
- Both lean ($\phi = 0.7$) and rich ($\phi = 1.3$) mixtures are well captured.

These results demonstrate that SL-GPS scales effectively to large mechanisms (> 500 species) while maintaining high accuracy and achieving dramatic mechanism reduction.